MAT201 - Fall 2022
Integration Review

This is not an assignment and will not be submitted for credit. Instead, this is meant to highlight integration techniques from *Calculus I & II* that will arise naturally while working in this course. It is best to try these problems using the same rules we follow for all course work in in MAT201. Please read over the Academic Integrity Policy and Standard of Work pages for more details.

Evaluate the following integrals.

1 $\displaystyle\int \cos^2 x\,dx$

2 $\displaystyle\int \frac{\sin^5 x}{\cos x}\,dx$

3 $\displaystyle\int_0^{\pi/4} \tan\theta\,d\theta$

4 $\displaystyle\int_{-\pi/3}^{\pi/3} \tan^4\theta\,d\theta$

5 $\displaystyle\int \frac{e^{1/x}}{x^2}\,dx$

6 $\displaystyle\int e^{\sqrt{x}}\,dx$

7 $\displaystyle\int_1^{e} \frac{dx}{x\ln^2 x + x}$

8 $\displaystyle\int_2^{4} \ln^2 x\,dx$

9 $\displaystyle\int \frac{dx}{\sqrt{4-x^2}}$

10 $\displaystyle\int \frac{dx}{(4-x^2)^{3/2}}$